

Problem Statement 2 – Output Guidelines

Package: MG029A – Acute Exacerbation of COPD

Purpose of the Outputs

Participants must generate **two outputs for each claim**:

S. No.	Output
1	Structured, row-wise output that captures: • Objective interpretation of the radiological image (Chest X-ray) • Correlation with written radiology report • Alignment with STG expectations • Stage and timeline awareness • Review-ready flags
2	Textual summary capturing all details of the structured output in a human readable format suitable for PPD / CPD review and must not contain diagnostic, approval, or payment decisions

✦ **Note:** For Acute exacerbation of COPD, only post-procedure imaging is expected. No re-procedure radiological image is mandated.

SECTION 1: Claim Details *(1 column)*

Claim No.

- **Description:** Unique identifier for the claim
- **Expected Content:** Numeric or alphanumeric claim identifier
- **Purpose:** Links image interpretation to a specific claim instance

SECTION 2: Image Classification *(1 column)*

Type of Radiological Image in Claim Submission / Post-Procedure Stage

- **Description:** Free text, Type of image submitted with the claim
- **Examples:**
 - “Chest X ray”, “Chest CT”, “Chest MRI” etc.
 - “Absent” (if no image submitted)
- **Purpose:** Confirms whether the mandatory imaging for COPD has been submitted

SECTION 3: Post-Procedure Image Description (Chest X-ray) *(5 columns)*

✓ Describe *what is seen*, not what it means

These fields capture **objective, observable image features described under following five mandatory headings/ columns -**

1. **Lung Fields:** free text, Overall appearance of lung parenchyma
 - Examples of features noted may include:
 - Clear
 - Hyperinflated
 - Patchy opacities
 - Basal opacities
 - Not clear (poor inspiration / artefacts)
2. **Left and Right Costophrenic (CP) Angles:** free text; Appearance of CP angles
 - Examples of features noted may include:
 - Sharp
 - Blunted (unilateral / bilateral)
 - Not assessable
3. **Left and Right Hilum:** free text; Appearance of hilar shadows
 - Examples of features noted may include:
 - Normal
 - Prominent

- Not well visualised
4. **Midline Shift:** free text; Position of trachea / mediastinum
 - Examples of features noted may include:
 - Not seen
 - Slight left / right shift
 5. **Cardiac Size:** free text; Visual impression of cardiac silhouette
 - Examples of features noted may include:
 - Normal
 - Enlarged
 - Appears small / narrow

SECTION 4: Report Validity and Correlation (only for post-procedure stage in this package) (4 columns)

1. **Intra-Report Consistency Check:** Whether the written report's observations and impression agree with each other; expected values:
 - Consistent
 - Partially supported
 - Unsupported
2. **Reasoning:** free text, examples may include:
 - "Observations match impression"
 - "Impression given without supporting observations"
3. **Image vs Written Report Correlation:** Whether image features support what the written report claims; expected values:
 - Consistent
 - Partially supported
 - Unsupported
4. **Reasoning:** free text, examples may include:
 - "X-ray features support reported findings"
 - "Image does not show features claimed in report"

SECTION 5: Claim, Package & STG Alignment (overall) (4 columns)

5A. Claim and Package Alignment: two columns to be filled -

1. **Alignment Category:** Expected values -
 - Both pre-procedure and post-procedure investigations align
 - Only pre-procedure investigations align
 - Only post-procedure investigations align
 - None of the investigations align
2. **Reasoning for alignment/ partial alignment/ nonalignment:** free text, examples may include:
 - “Only post-procedure image is required as per STG”
 - “Pre-procedure X-ray submitted but STG expects post-procedure image”

5B. Investigation and STG Alignment: two columns to be filled -

3. **STG Alignment Outcome:** Expected values -
 - Consistent
 - Partially supported
 - Unsupported
4. **Reasoning for alignment/ partial alignment/ nonalignment:** Textual explanation mapped to PPD / CPD logic of STGs; examples may include:
 - “Hyperinflation seen; other features of acute exacerbation not evident”
 - “Image quality limits assessment”

SECTION 6: Stage & Timeline Awareness (5 columns)

S. No.	Column	Description
1	Admission Date	DD/MM/YYYY
2	Pre-Procedure Investigation Date	DD/MM/YYYY
3	Post-Procedure Investigation Date	DD/MM/YYYY
4	Discharge Date	DD/MM/YYYY
5	Flag before admission date or after discharge date documents for post-procedure investigations	Yes / No

SECTION 7: Quality (1 column)

- **Description:** free text, image usability comment, examples may include:
 - Good enough for human eyes
 - Poor quality, artefacts present

- Overexposed / underexposed

Summary PDF

- **Description:** Short human-readable case summary
- **Expected Content:**
 - Key image description summary
 - Report validity and correlation outcome
 - Package and STG alignment statement
 - Stage and timelines awareness summary of dates and flags
- **Tone:** Assistive, neutral, non-diagnostic

✔ Important Rules for Participants

- Use **observational language only**
- Do **not** confirm diagnoses (e.g., pneumonia, COPD severity)
- Do **not** recommend approval or rejection
- Explicitly mention uncertainty if image quality is poor
- All fields must be populated (use “Not assessed / Not seen” where needed)